

# Intergenerational Housing Misallocation and Fertility

## Preliminary Note

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### Abstract

Children increase a household's demand for living space. Yet large rental units are scarce, buying a family-sized home requires liquid wealth and access to credit, and older households own much of the stock of large homes. I study whether this allocation of housing raises the cost of having children for young households. I first develop a simple overlapping-generations model in which rental limits, down-payment constraints, and the retention decisions of older owners determine who occupies family-sized housing. I then study the same forces in a finite-life model with income risk, saving, tenure choice, transaction costs, and bequests. The model distinguishes changes in family size among current residents from changes in local births due to entry into the housing market. The preliminary calibration suggests that restricted access to family-sized housing matters for households deciding whether and when to have children, and that home purchases and births are closely connected near the end of the fertile years.

### Status of this draft

This is a preliminary note. The analytical results below state the assumptions needed for each derivation. The quantitative results use the July 9, 2026 calibration and should not yet be interpreted as final estimates. I have since corrected the timing with which current housing choices enter market demand and the moments used in estimation. Because the correction changes several target moments, the model must be re-estimated before the policy magnitudes can be evaluated.

The specification below matches the one used for the July estimates: the owner premium applies to usable housing space, bequest utility varies with family size, and all owners face the same debt limit. I also recovered the income profile and wealth grid used in July. Restoring these settings improves reproducibility but does not change the estimates reported in this note. All policy results will be recomputed after re-estimation.

## 1 Introduction

A child changes a household's housing problem. It is not only another set of goods to buy, or more time to spend at home. It is often a need for a different dwelling: another bedroom, a larger living space, or a unit where raising a child is feasible. In many high-cost cities this margin is hard to adjust. Large rental units are scarce, family-sized homes are concentrated in ownership, and ownership requires down payments and borrowing capacity at ages when many households have little wealth.

This paper studies whether those space and financial constraints affect final family size and the number of births generated by a cohort. That distinction matters. A housing shock can move births across years without changing completed fertility. It can also change local births by changing who enters a high-cost housing market, even if completed fertility among incumbent households moves little. The question here is when housing constraints change completed fertility, when they mainly change entry and local scale, and how both effects depend on who can access family-sized space over the lifecycle.

The mechanism has three parts. First, children raise the household’s demand for space. If the rental market does not offer sufficiently large units, the shadow price of an additional room can exceed the posted rent. Second, the natural substitute is ownership, but buying a family-sized home requires liquidity and credit. A young household may value the larger unit and still be unable to purchase it. Third, older owners may remain in large homes even when their own marginal value of space is low, because transaction costs and tax rules subsidize staying put. Housing scarcity then shows up not only as a high price of space, but also as a mismatch between who occupies family-sized units and who values them for fertility.

I begin with a compact two-period model. It shows how rental limits, down-payment constraints, and old-owner retention raise the value of family-sized space for constrained young households and, through that channel, raise the cost of having children. I then embed the same economics in a finite-life quantitative model. The model combines lifecycle fertility choice (Sommer, 2016; Doepke and Kindermann, 2019) with housing tenure choice, collateral constraints, and owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The analytical housing model follows the one-market floorspace logic in Coven et al. (2025): all housing services clear in one market, while tenure, borrowing, and tax rules determine who can actually occupy family-sized units.

I use the current calibration for two purposes. First, I ask whether the model places a high value on family-sized space for households deciding whether to have children. Second, I conduct a preliminary study of housing policies that could move large homes toward younger, constrained households. The calibration finds many renters at the size limit when they are choosing family size and produces a close connection between births and purchases near the end of the fertile window. It also understates early ownership, overstates young liquid wealth, and produces too little exit from ownership in old age. I therefore use these results to organize the evidence and the next estimation step, not to make a settled quantitative claim.

## 2 Related Literature

The paper connects work on fertility choice to work on housing tenure and the allocation of large homes. Since Becker (1960) and Becker and Lewis (1973), economics has treated children as choice variables that enter preferences and require resources. Quantitative versions place that choice in lifecycle environments with income risk, saving, timing, and intrahousehold bargaining (Jones et al., 2010; Sommer, 2016; Doepke and Kindermann, 2019). Recent work emphasizes that the U.S. fertility decline is difficult to explain with one postponement margin or period shock (Kearney et al., 2022).

Housing prices, housing wealth, and mortgage finance also move fertility behavior. Dettling and Kearney (2014) find opposite house-price effects for owners and nonowners, Lovenheim and Mumford (2013) find that housing wealth raises homeowners’ fertility, and Cumming and Dettling (2024) show that mortgage-rate pass-through affects births through cash flow. Hacamo (2021) and Dettling

and Kearney (2025) connect mortgage access to home purchases and childbearing. In equilibrium, Couillard (2025) show that the supply of large units matters for aggregate fertility.

The outcome here is closer to completed fertility than to a current birth rate. Simon and Tamura (2009) find that rents per room predict lower fertility, including in completed-fertility specifications. Clark (2012) provides an important caution: expensive housing markets delay first births but do not clearly reduce completed fertility. More recent evidence links housing access to both timing and final family size (Japaridze and Sayour, 2024; Dettling and Kearney, 2025; van Doornik et al., 2025). I use this evidence to distinguish current births, completed fertility among incumbent households, and cohort births after housing-market entry responds.<sup>1</sup>

The closest housing-allocation paper is Coven et al. (2025). Their model studies how property taxes, capitalization, and financial constraints allocate housing between older homeowners and younger constrained households. I add children to that allocation problem. Family-sized space held by an incumbent can then change the private price of children for a young household. The retention force also relates to mortgage lock-in and housing ladders (Fonseca and Liu, 2023; Fonseca et al., 2026). My baseline quantitative model uses transaction costs rather than fixed-rate mortgage lock-in, but the allocation logic is similar.

The quantitative housing model builds on work in which tenure, collateral constraints, and owner-occupied housing shape household behavior (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). It also relates to spatial work in which housing supply and local prices allocate households across markets (Rosen, 1979; Roback, 1982; Gyourko et al., 2013; Hsieh and Moretti, 2019). The present model keeps one housing market so that tenure, family size, and the allocation of space remain the central margins.

### 3 Analytical Model

#### 3.1 Environment

The economy is populated by overlapping generations of young and old households. A young household has type  $i = (y_i, b_i)$ : income and wealth at entry. Wealth is inherited from the previous old generation<sup>2</sup>. A household decides whether to enter the economy; inside, she chooses a tenure (rent or own), a house size, completed fertility, and savings for old age. Old households, indexed by  $j$ , are of two kinds. Old incumbents, last period’s young owners, who enter old age with financial wealth, the house they bought, and a possible accrued gain, choose how much of the house to keep and how large an estate to leave, then exit. The other type is old renters, last period’s young renter entrants, who rent in old age and carry no wedge. Nonentrants stay outside the market in both periods. Each period a mass  $\bar{M}$  of potential young households enters the pool: the maturing children of the previous cohort plus a renewal inflow. Types are distributed according to an exogenous  $G_Y$ .

Households have preferences over consumption, housing, and children. Children raise the household’s consumption and housing needs: each child claims  $\chi > 0$  units of consumption and  $\kappa > 0$  units of space, so with consumption spending  $c_i$ , housing  $h_i$ , and fertility  $n_i$ ,

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

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<sup>1</sup>Life-course evidence documents that fertility, homeownership, dwelling size, and residential transitions move together (Mulder and Billari, 2010; Kulu and Steele, 2013; Chudnovskaya, 2019). The model treats those joint transitions as economic choices over the lifecycle.

<sup>2</sup>Distributed randomly, without a tie to bequests and inheritants. It can be distributed unequally, however.

Old households value consumption, housing, and the bequest  $e_j \geq 0$  they leave at exit:

$$u^2 = \log c_j^2 + \gamma \log h_j^2 + \mathcal{B}(e_j),$$

with  $\mathcal{B}$  a warm-glow function, increasing and concave, possibly zero, common to all old households. The exiting cohort's estates are what the arriving cohort inherits.

There is a fixed stock  $\bar{H}$  of divisible floorspace, held by old incumbents and by passive absentee landlords: buyers purchase from either, renters occupy landlord space, and rents and sale proceeds are transfers.<sup>3</sup> Renting and owning are occupancy modes with different size limits, the largest sizes available to rent and to own,

$$h \leq \bar{h}^R \text{ (rent)}, \quad h \leq \bar{h}^O \text{ (own)}, \quad \bar{h}^R < \bar{h}^O,$$

Family-sized houses are mostly available to buy, not to rent. All floorspace trades in one market: the asset price is  $P$ , and no-arbitrage ties the rental rate to it,

$$q = \rho P, \quad \rho = r + \delta + \tau^P.$$

The renter pays  $q$  as rent; the owner pays it as mortgage interest, property tax, depreciation, and the forgone return on her equity—different payments, equal per-unit value by arbitrage. Given  $q$ , a higher property tax lowers  $P$  and with it the down payment on any given house.

A buyer finances at most a share  $\phi \in (0, 1)$  of the purchase, so she posts a down payment  $(1 - \phi)Ph$  at purchase, out of her liquid wealth. Households can save in an exogenously provided bond, but renters cannot borrow. The interest rate  $r$  is exogenous.

Two taxes appear: the property tax  $\tau^P$ , part of the user cost above, and the capital-gains tax. Selling a house triggers a tax: the seller pays the rate  $\tau^g$  on the difference between the sale price and what she paid, in dollars. Dying instead passes the house with a stepped-up basis - the heir's purchase price is reset to the current value - so the gain is never taxed<sup>4</sup>.

Where do gains come from when the real price never moves? From inflation. Dollar prices and incomes all grow at rate  $\varpi$ , so relative prices and real choices are unaffected; but the tax form compares dollars from different dates, and a house bought one period ago shows a paper gain even though its real value is unchanged. The tax on that paper gain is real,

$$\bar{\ell} = \tau^g \frac{P \varpi}{1 + \varpi}$$

per unit sold, and this is the model's only contact with inflation. Every cohort carries the same gain, so every incumbent's unit bears  $\bar{\ell}$  if sold while alive.<sup>5</sup> An estate tax, by contrast, would fall on the transfer in any form and leave the sell-or-keep margin undistorted; the gains tax distorts it precisely because death forgives it. Everything is deterministic: a buyer knows the tax her future self will face<sup>6</sup>.

<sup>3</sup>There is no construction: the model allocates a fixed stock. This goes together with the stationary equilibrium focus. More sophisticated supply will follow in the extended model and the quantitative application.

<sup>4</sup>This was the essence of the point suggested by Simon. I will go a bit deeper into it in the next weeks, but it is one of the two key margins (the other being financial constraints). The idea is then: agents have an incentive to hold on to the house, because this way neither they nor their heirs pay taxes.

<sup>5</sup>The statutory exclusion, which exempts gains below a dollar threshold, is an institutional detail I suppress; with it, small holdings escape the tax and lock-in sorts by size.

<sup>6</sup>So I am resorting to a trick here to have a capital gain in a stationary model. [Coven et al. \(2025\)](#) have a real growth term there, but they don't have my problem with fertility. More thought is required on this.

### 3.2 Households

Conditional on entry, household  $i$  chooses a tenure mode  $m \in \{R, O\}$ , housing, fertility, consumption and savings; old age is discounted at  $\beta$ . For  $m \in \{R, O\}$ ,

$$W_i^m = \max_{c_i, h_i, n_i, b'_i} \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ + \beta [\mathbf{1}\{m = R\} V^R(b_{j'}) + \mathbf{1}\{m = O\} V^O(b_{j'}, h_i)]$$

subject to

$$c_i + q h_i + b'_i = y_i + b_i, \quad h_i \leq \bar{h}^m, \quad b'_i \geq 0, \\ \mathbf{1}\{m = O\} (1 - \phi) P h_i \leq b_i.$$

Financial wealth carried into old age is  $b_{j'} = (1 + r) b'_i - \mathbf{1}\{m = O\} q h_i$ . Children have a market price: each child claims  $\chi$  goods and  $\kappa$  units of space, and space rents at  $q$ , so providing for a child costs  $\chi + \kappa q$  per period. A useful benchmark is equal scaling,  $\chi = \kappa q / \alpha$ : the child splits her cost between space and goods in the proportion  $\kappa q / \chi = \alpha$ , the same proportion in which her parents split their own spending.

The old incumbent enters with financial wealth  $b_j$ , housing  $H_j^0$ , and an accrued gain; old households have no income. She faces a choice with respect to her housing: sell now and pay the tax, or keep it until death, when the tax is forgiven.<sup>7</sup> She solves

$$V^O(b_j, H_j^0) = \max_{c_j^2, h_j^2, e_j} u^2(c_j^2, h_j^2, e_j) \\ \text{subject to } c_j^2 + e_j + q h_j^2 = b_j + q H_j^0 - \bar{\ell} (H_j^0 - h_j^2), \quad 0 \leq h_j^2 \leq H_j^0.$$

Selling a unit nets  $q - \bar{\ell}$ ; hence the stepped-up basis acts as a subsidy to locking in. An old renter has no carried housing and no tax:

$$V^R(b_j) = \max_{c_j^2, h_j^2, e_j} u^2(c_j^2, h_j^2, e_j) \quad \text{subject to } c_j^2 + e_j + q h_j^2 = b_j, \quad 0 < h_j^2 \leq \bar{h}^R,$$

and at an interior choice her marginal value of space is exactly  $q$ .

Using  $P = q / \rho$ , owner housing is bounded by

$$H_i^O = \min \left\{ \bar{h}^O, \frac{b_i \rho}{(1 - \phi) q} \right\}, \quad H_i^R = \bar{h}^R.$$

$H_i^m$  is the effective family-housing cap: a renter is capped by the stock, a buyer by the size limit or by collateral, whichever is shorter. Tenure is  $W_i^Y = \max\{W_i^R, W_i^O\}$ . Entry compares the inside value with an outside option  $\bar{W}^E$  under extreme-value shocks of scale  $\kappa_E$ :

$$\pi_i^E = \frac{\exp\{W_i^Y / \kappa_E\}}{\exp\{W_i^Y / \kappa_E\} + \exp\{\bar{W}^E / \kappa_E\}}$$

is the probability that household  $i$  enters.

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<sup>7</sup>In the more realistic model we will have indivisible housing, so this will look different. With an indivisible house the choice is all-or-nothing: a strict keeper escapes the bill entirely, lock-in concentrates into a notch at the downsizing margin, and pre-saving becomes plan-contingent.

### 3.3 Equilibrium

Aggregate demand adds young households, old incumbents, and old renters, and the single market clears:

$$\bar{M} \int \pi_i^E h_i dG_Y(i) + \int h_j^2 dF_O(j) + \int h_j^2 dF_R(j) = \bar{H},$$

where  $F_O$  and  $F_R$  are the measures of old incumbents and old renters. A stationary equilibrium consists of a user cost  $q$  (and  $P = q/\rho$ ), young policies and entry probabilities, old policies, and stationary measures  $(\bar{M}, G_Y, F_O, F_R)$  such that:

1. the policies solve the household problems above at  $q$ ;
2. the floorspace market clears, as displayed;
3. the population reproduces itself: young owners become the old incumbents, young renters the old renters, and births plus the renewal inflow recreate the cohort;

### 3.4 Equilibrium policies

The Lagrangian for a young agent is

$$\begin{aligned} \mathcal{L} = & u_i^Y + \beta V_{j'} + \lambda_i (y_i + b_i - c_i - qh_i - b'_i) \\ & + \mu_i (b_i - (1 - \phi)Ph_i) + \theta_i (\bar{h}^m - h_i) + \xi_i b'_i, \end{aligned}$$

with  $\mu_i = 0$  in the rental mode and  $\xi_i \geq 0$  the multiplier on the no-borrowing bound  $b'_i \geq 0$  for a renter. The formulas below apply at the stated interior saving and retention margins; at a corner, the corresponding complementary-slackness condition replaces the equality.

The first-order condition for saving sets the marginal value of resources today against the return tomorrow:

$$\frac{1}{c_i - \chi n_i} = \beta(1 + r) \frac{1}{c_{j'}^2} + \xi_i, \quad \xi_i b'_i = 0,$$

which for savers ( $\xi_i = 0$ ), under logarithmic old utility (warm glow  $\omega_B \log e$  included) and interior old choices, collapses to a savings rule:

$$b'_i = \beta(1 + \gamma + \omega_B)(c_i - \chi n_i) + \mathbf{1}\{m = O\} \frac{\bar{\ell} h_i}{1 + r},$$

with  $\gamma$  the old housing weight and  $\omega_B$  the warm-glow weight: the household saves old age's claim on adult consumption— $\Lambda \equiv \beta(1 + \gamma + \omega_B)$ —plus her discounted future tax bill, so anticipating the lock-in tax is itself a savings motive. By the same conversion, a buyer pays an effective price  $q^O = q + \bar{\ell}/(1 + r)$  per unit of space—the market price plus her own discounted lock-in tax, pre-funded through savings rather than paid as a flow—while a renter pays  $q$ ; write  $q^m$  for the mode's effective price. A unit of adult consumption commits  $1 + \Lambda$  units of resources, one spent now and  $\Lambda$  carried.

Then, with no binding constraints, with lifetime resources  $w_i = y_i + b_i$ ,

$$c_i - \chi n_i = \frac{w_i}{1 + \alpha + \vartheta + \Lambda}, \quad n_i = \frac{\vartheta w_i}{(1 + \alpha + \vartheta + \Lambda)(\chi + \kappa q^m)}, \quad h_i = \frac{\alpha w_i}{(1 + \alpha + \vartheta + \Lambda) q^m} + \kappa n_i,$$

Dividing the housing derivative of  $\mathcal{L}$  by the consumption derivative,  $\lambda_i = 1/(c_i - \chi n_i)$ , prices every multiplier in goods. For a renter this gives

$$\frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \quad \zeta_i^R = \frac{\theta_i}{\lambda_i} \geq 0:$$

the left side is  $u_h/u_c$ , her marginal rate of substitution between space and consumption—the goods she would give up for one more unit of space, holding utility fixed—and it exceeds the rent exactly by the bite of the rental size limit, computable from her observed bundle. For a young owner the same two derivatives give

$$\frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O, \quad \zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i}(1 - \phi)P}_{\zeta_i^{O,F}, \text{ down payment}} + \underbrace{\frac{\theta_i}{\lambda_i}}_{\text{size limit}} :$$

her marginal value of space exceeds her effective price by the bite of whichever access constraints bind—the down payment ( $\mu_i$ ) or the owner size limit ( $\theta_i$ ). When the size limit is slack, the bundle identifies the financing bite alone ( $F$  for financing):  $\zeta_i^{O,F}$  is the observed ratio minus  $q^O$ . For an old incumbent at an interior retention choice,

$$\frac{\gamma c_j^2}{h_j^2} = q - \bar{\ell} :$$

she keeps marginal space she values below what the market would pay, because selling it would realize the taxable gain. Old renters carry no such wedge: their marginal value is  $q$ .

The solved policies give the level of fertility; the first-order condition behind them shows where housing enters it. Dividing that condition by the marginal utility of consumption prices the marginal child in goods: in either tenure,

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}.$$

A child's space requirement is priced at the household's own shadow value of space, so a binding access constraint acts as a tax of  $\kappa \zeta_i^m$  per child. Children get more expensive exactly where family-sized space is hard to reach, with market prices unchanged.

### 3.5 Efficiency

I first ask whether the housing frictions described above can produce an inefficient allocation of family-sized space. To isolate this question, I consider a planner that faces the housing stock, size limits, goods feasibility, and the domain restrictions, but can use lump-sum transfers to support an allocation. The planner therefore does not face private borrowing constraints, which restrict purchases rather than the physical feasibility of occupying a home.

**Proposition 1.** *Hold the stationary cross-section, entry, tenure assignments, and savings fixed; the variation below resizes two owner-occupied houses and converts no tenures. The planner's surplus from moving one unit of space from an old incumbent at an interior retention choice to a young owner whose owner size limit is slack and whose financing constraint binds ( $\zeta_i^{O,F} > 0$ ) is*

$$\underbrace{(q^O + \zeta_i^{O,F})}_{\text{buyer's marginal value}} - \underbrace{(q - \bar{\ell})}_{\text{incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} > 0.$$

*If this compensated local reassignment is feasible at zero real implementation cost and the displayed surplus is positive, the competitive allocation does not solve the planner's conditional allocation problem.*

The proposition identifies a possible gain from moving space from an old owner to a constrained young buyer. The buyer values another unit at her effective price plus the financing gap, while the incumbent values the unit at the net-of-tax sale price. The market price cancels in the difference, and the tax payments are transfers rather than resource costs. This does not imply that every housing subsidy is welfare improving. Any real cost of relaxing the borrowing constraint must be deducted from the surplus, and a constraint that policy cannot relax creates no implementable gain.

Figure 1 illustrates this comparison. The buyer's marginal value of space lies above her effective price by  $\zeta^{O,F}$ , while the incumbent's value lies below the market price by  $\bar{\ell}$ . Compensating the incumbent at  $q - \bar{\ell}$  and charging the buyer  $q^O + \zeta^{O,F}$  leaves both households as well off as before and generates  $\zeta^{O,F} + \bar{\ell}/(1+r) + \bar{\ell}$  in surplus.

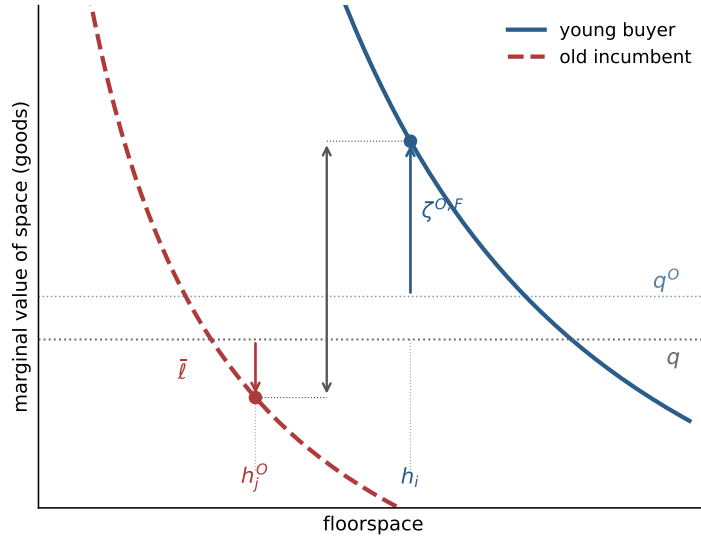


Figure 1: Misallocation. The collateral-capped buyer values marginal space at  $q^O + \zeta^{O,F}$ , above the market price; the locked-in incumbent values it at  $q - \bar{\ell}$ , below; the distance between the two points is the recoverable surplus.

### 3.6 Fertility and policy

Fix a tenure mode and a household whose cap binds, so that her housing is pinned at the effective family-housing cap defined in the household problem: the rental size limit for a renter, the shorter of the owner size limit and the collateral bound for a buyer. Write  $H$  for the level of that cap,  $w = y + b$  for lifetime resources, and  $q^m$  for the mode's effective price. We relax the cap by a unit of space and study fertility. Concentrating the savings rule into the budget, fertility at the cap solves  $\vartheta/n = (1 + \Lambda)\chi/(w - q^m H - \chi n) + \alpha\kappa/(H - \kappa n)$ ; differentiating with respect to the cap level,

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{(H - \kappa n)^2} - \frac{(1 + \Lambda)\chi q^m}{(w - q^m H - \chi n)^2}}{\frac{\vartheta}{n^2} + \frac{(1 + \Lambda)\chi^2}{(w - q^m H - \chi n)^2} + \frac{\alpha\kappa^2}{(H - \kappa n)^2}},$$

which is positive for every binding cap if and only if  $\chi \leq (1 + \Lambda)\kappa q^m/\alpha$ : relaxing access raises fertility whenever the child's consumption requirement does not exceed its space requirement priced



at the lifetime cost of adult consumption. The derivative races two forces: extra space makes the child's space floor  $\kappa$  easier to meet, pushing fertility up, while paying  $q^m$  for the space tightens the budget around the child's goods floor  $\chi$ , pushing fertility down. The savings margin weakens the second force: the housing bill is paid partly by saving less, so only a fraction  $1/(1 + \Lambda)$  of it falls on current spending, and the threshold below which access raises fertility scales up by  $(1 + \Lambda)$ . Figure 2 draws the relation: fertility rises along the binding cap at rate  $dn/dH$  and flattens at  $h^u$ , where the cap stops binding.

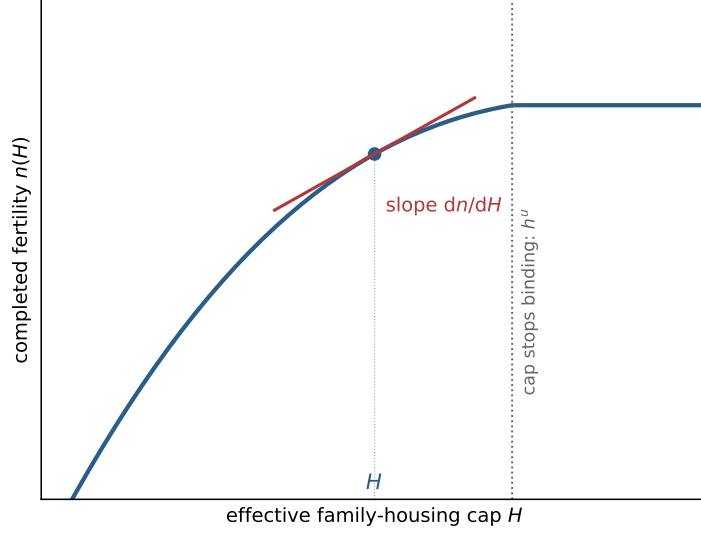


Figure 2: The fertility margin. Completed fertility against the effective family-housing cap, from the same solved case as Figure 1; the slope at the equilibrium cap is  $dn/dH$ , and the cap stops binding at  $h^u$ .

I next ask how a housing policy that relaxes these constraints affects family size. The aim is to improve the allocation of family-sized space, not to target fertility directly. Fertility changes only because the policy changes the cost of accommodating children. A down-payment grant raises  $b_i$ , looser credit raises  $\phi$ , and either change relaxes the collateral bound  $b_i\rho/((1 - \phi)q)$ . Making larger units available to rent raises  $\bar{h}^R$ . These policies leave the aggregate stock  $\bar{H}$  unchanged and change who can occupy it. Let  $\bar{n}_i^E$  denote the number of children in the outside option, normalized to zero in the baseline. At fixed prices, the effect of a small policy  $x$  on aggregate fertility is

$$\left. \frac{dN}{dx} \right|_q = \bar{M} \sum_m \int_{\mathcal{E}_m(x)} \left[ \pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E) \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m \right] \mathcal{P}_i^m(x) dG_Y(i),$$

where  $\mathcal{E}_m(x)$  is the set of households whose cap binds in mode  $m$ ,  $\mathcal{P}_i^m = \partial H_i^m / \partial x$  is the pass-through of the policy to the binding component of the cap,  $\Psi_i^m = dn/dH$  is the fertility response, and  $\Theta_i^m = \zeta_i^m / (c_i^m - \chi n_i^m)$  drives the entry term. A policy aimed at a slack constraint moves nothing: a down-payment grant does nothing for a household pinned by the rental size limit. At  $\chi = (1 + \Lambda)\kappa q^m / \alpha$ , equal scaling at lifetime prices,  $\Psi_i^m$  is proportional to  $\zeta_i^m / (c_i^m - \chi n_i^m)$ , so the same gap that measures misallocation predicts whose fertility responds.

Policies can also change tenure. A change in  $W_i^O - W_i^R$  moves some households across the rent-own boundary, and the gains tax affects this choice because a buyer anticipates the tax due upon a future sale.

## 4 Quantitative Lifecycle Model

### 4.1 Lifecycle and Income

The quantitative model embeds the same mechanism in a finite-life economy. It has the standard lifecycle structure: households age, face income risk, save, choose whether to rent or own, choose housing size, decide fertility, and eventually leave bequests. The housing market has one aggregate user cost and one asset price. Tenure matters because it changes the size menu, financing constraints, transaction costs, and tax exposure.

Time is discrete. Households enter adult life at age  $a = 1$ , retire after age  $J_R$ , and die at the end of age  $J$ . Before retirement, earnings depend on age and idiosyncratic productivity  $z \in \mathcal{Z}$ . Income is

$$y(a, z) = \begin{cases} (1 - \tau_{\text{pay}})we_a z, & a \leq J_R, \\ p^{\text{ret}}, & a > J_R, \end{cases} \quad z' \sim \Pi^z(\cdot|z), \quad (1)$$

The wage is common apart from age and productivity because the quantitative mechanism is housing access, tenure, and lifecycle wealth.

### 4.2 Preferences and Bequests

Let  $n_s$  be the number of dependent children living at home. Completed family size  $n$  and the current number at home differ because children eventually mature. Household needs are

$$\begin{aligned} \bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n n_s, \\ \bar{h}(n, s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s. \end{aligned} \quad (2)$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Flow utility is Stone–Geary over consumption and usable housing space, wrapped in CRRA. Write  $h^R$  for rented space and  $h$  for the owned housing stock; only one can be positive. The owner service premium applies after household needs are removed:

$$u(c, h^R, h; n, s) = \frac{[(c - \bar{c}(n, s))^{\alpha_c} \mathfrak{h}(h^R, h; n, s)^{1-\alpha_c}]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} n_s, \quad (3)$$

where

$$\mathfrak{h}(h^R, h; n, s) = \begin{cases} h^R - \bar{h}(n, s), & h = 0, \\ \chi_O [h - \bar{h}(n, s)], & h > 0, \end{cases} \quad \chi_O > 0. \quad (4)$$

Utility is  $-\infty$  when residual consumption or usable space is nonpositive. In market clearing, an owner demands the physical quantity  $H_k$ , not the preference-scaled quantity  $\chi_O[H_k - \bar{h}(n, s)]$ .

At death, households receive warm-glow bequest utility. The house passes to the estate at market value: there is no forced sale, no transaction cost, and no tax at death, so bequeathable wealth is  $W^B = b + Ph$ , liquid wealth plus the gross value of any house held. Bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0(1 + \theta_n n) \frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma}}{1 - \sigma}. \quad (5)$$

This utility is not normalized to zero at a zero estate. Because its scale depends on  $n$ , that level is part of the family-size choice. The parameter  $\theta_n$  lets completed fertility affect the desired estate even after children have left the household.

### 4.3 Housing, Finance, and Taxes

There is one aggregate housing-services market. The user cost  $q$  and asset price  $P$  satisfy

$$q = (r + \delta + \tau^P)P. \quad (6)$$

Housing supply is upward sloping:

$$H^S(q) = \bar{H} \left( \frac{q}{\bar{q}} \right)^\eta. \quad (7)$$

Here  $\bar{q}$  is the reference user cost, so housing supply equals  $\bar{H}$  at  $q = \bar{q}$ . The fixed-stock compact model is the limiting case  $\eta = 0$  around the benchmark stock. More generally,  $\eta$  determines whether an increase in housing demand appears mainly as a higher user cost or as additional housing.

Households either rent housing services or own one of  $K$  discrete housing sizes. The owned housing stock  $h$  belongs to

$$\mathcal{H} = \{0, H_1, \dots, H_K\}, \quad (8)$$

where  $h = 0$  denotes a renter and  $h = H_k$  an owner of a house of size  $H_k$ , ordered so that  $H_1 < \dots < H_K$ . Thus tenure is implicit in the housing position rather than carried by a separate indicator. Renters choose  $h^R$  continuously up to  $\bar{h}^R$ ; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Section 3.

Renters choose  $h^R \in [0, \bar{h}^R]$  and cannot borrow. Owners choose from the discrete ladder and can borrow against the house. They pay maintenance and property taxes through  $(\delta + \tau^P)PH_k$ .

At origination, a buyer can finance at most a share  $\phi$  of house value, so the required down payment is

$$(1 - \phi)PH_k. \quad (9)$$

Equivalently, adjusted liquid wealth immediately after purchase must satisfy  $\tilde{b} \geq -\phi PH_k$ . End-of-period saving for every owner, whether a buyer or an incumbent, faces the same debt floor:

$$b' \geq -\phi PH_k. \quad (10)$$

The down payment governs entry into ownership, while the debt floor governs resources after purchase. The distinction matters because a household can be able to service an owned home and still lack the liquid wealth needed to buy it.

Housing adjustment is costly. A sale of an owned house  $h > 0$  incurs the proportional transaction cost  $\psi^s$ , so net sale proceeds are

$$\Omega(h) = (1 - \psi^s)Ph. \quad (11)$$

Transaction costs are the only disposal friction in the computed benchmark: an owner who sells pays  $\psi^s$  per unit of value and faces no tax on accrued gains.

The capital-gains treatment of Section 3 is specified as an extension and is not part of the computed benchmark. Pricing realized gains requires tracking a tax basis  $B^p$  for each owner; a sale would then trigger  $T^g(PH_k, B^p) = \tau^g \max\{PH_k - B^p - \bar{g}, 0\}$  with statutory exclusion  $\bar{g}$ , net proceeds would fall to  $\Omega(H_k) - T^g(PH_k, B^p)$ , and stepped-up basis at death would forgive the accrued gain. The benchmark does not track  $B^p$ : the old-owner lock-in wedge  $\bar{\ell}$  is priced analytically in Section 3, and the counterfactual section instead studies a direct change in old-owner retention.

#### 4.4 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \dots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \dots, \bar{n}\}. \quad (12)$$

The option  $n' = 0$  means no birth in the current period. A positive choice fixes completed fertility forever and moves its children into the first dependent stage. A household that reaches the end of the fertile window without a positive choice remains childless.

In the computation,  $\mathcal{N} = \{0, 1, 2\}$  and the last element is the open-ended highest family-size category. The completed-fertility moment matched to the CPS is reported as  $2\mathbb{E}[n]$ . This household-to-woman conversion is applied only to that reported moment. Child needs and bequest utility use the encoded family-size category  $n$ , while the housing-size target maps  $n = 1$  to the 1–2 group and  $n = 2$  to the 3+ group. The distinction is a unit convention and does not create an additional fertility choice.

Let  $s = 0$  denote no dependent children and let  $s = 1, \dots, S_c$  denote dependent-child stages. After a positive fertility choice, the household enters  $s = 1$ . Children age stochastically through the stages according to  $\Pi^c$ . While children are at home,  $n_s = n$ ; after maturity,  $n_s = 0$ . Completed fertility still matters through bequest utility.

#### 4.5 Household Problem

Within the period, the household first faces fertility if eligible, then chooses its owned housing stock  $h'$  or rental services  $h^{R'}$ , and then chooses consumption and saving. A renter sets  $h' = 0$ ; an owner sets  $h^{R'} = 0$ . Let the continuation value after income risk and child aging be

$$\bar{V}(b', z, h', a + 1, n, s) = \sum_{z'} \sum_{s'} \Pi^z(z'|z) \Pi^c(s'|s, n) V(b', z', h', a + 1, n, s'). \quad (13)$$

The terminal value is

$$V(b, z, h, J + 1, n, s) = \mathcal{B}(b + Ph, n).$$

Conditional on renting, the branch value is

$$\begin{aligned} V^R(\tilde{b}, z, a, n, s) &= \max_{c, h^{R'}, b'} u(c, h^{R'}, 0; n, s) + \beta \bar{V}(b', z, 0, a + 1, n, s) \\ \text{s.t.} \quad &c + qh^{R'} + b' = (1 + r)\tilde{b} + y(a, z), \\ &0 \leq h^{R'} \leq \bar{h}^R, \quad b' \geq 0. \end{aligned} \quad (14)$$

Conditional on choosing an owned house  $h' > 0$  after any adjustment, the branch value is

$$\begin{aligned} V^O(\tilde{b}, z, h', a, n, s) &= \max_{c, b'} u(c, 0, h'; n, s) + \beta \bar{V}(b', z, h', a + 1, n, s) \\ \text{s.t.} \quad &c + (\delta + \tau^p)Ph' + b' = (1 + r)\tilde{b} + y(a, z), \\ &b' \geq -\phi Ph'. \end{aligned} \quad (15)$$

The debt floor is the same for a buyer and an incumbent. Purchase feasibility is screened separately through the down-payment condition before this branch is evaluated. Renting offers a continuous

choice of space but no borrowing; ownership offers the larger discrete menu and borrowing against the house, at the cost of the initial down payment and future adjustment costs.

The liquid wealth available after changing the owned housing stock from  $h$  to  $h'$  is

$$\tilde{b}(b, h, h') = b + \mathbf{1}\{h > 0, h' \neq h\}\Omega(h) - \mathbf{1}\{h' > 0, h' \neq h\}Ph'. \quad (16)$$

A seller collects net proceeds, a buyer pays the full purchase price, and a household that keeps its position pays nothing.

The housing-position value for a feasible  $h' \in \mathcal{H}$  is

$$\tilde{V}^H(h'|b, z, h, a, n, s) = \begin{cases} V^R(\tilde{b}(b, h, 0), z, a, n, s), & h' = 0, \\ V^O(\tilde{b}(b, h, h'), z, h', a, n, s), & h' \in \mathcal{H} \setminus \{0\}. \end{cases} \quad (17)$$

An infeasible branch is assigned value  $-\infty$ . The rent-or-own and owner-size choices have Type-I extreme-value shocks with scale  $\kappa_T$ :

$$\pi^H(h'|b, z, h, a, n, s) = \frac{\exp\{\tilde{V}^H(h'|b, z, h, a, n, s)/\kappa_T\}}{\sum_{\hat{h} \in \mathcal{H}} \exp\{\tilde{V}^H(\hat{h}|b, z, h, a, n, s)/\kappa_T\}}, \quad (18)$$

and the inclusive housing value is

$$V^H(b, z, h, a, n, s) = \kappa_T \log \sum_{\hat{h} \in \mathcal{H}} \exp\{\tilde{V}^H(\hat{h}|b, z, h, a, n, s)/\kappa_T\}. \quad (19)$$

At the estimated value  $\kappa_T = 0$ , these expressions mean their deterministic limit: the branch with the highest value is chosen with probability one, and a tie is resolved by the fixed ordering of feasible branches.

For a childless household in the fertile window, the value of fertility option  $n'$  is

$$\tilde{V}^F(n'|b, z, h, a, 0, 0) = V^H(b, z, h, a, n', s(n')), \quad s(n') = \mathbf{1}\{n' > 0\}. \quad (20)$$

With Type-I extreme-value shocks of scale  $\kappa_F$ ,

$$\pi^F(n'|b, z, h, a, 0, 0) = \frac{\exp\{\tilde{V}^F(n'|b, z, h, a, 0, 0)/\kappa_F\}}{\sum_{\hat{n} \in \mathcal{N}} \exp\{\tilde{V}^F(\hat{n}|b, z, h, a, 0, 0)/\kappa_F\}}. \quad (21)$$

The value before the fertility draw is the corresponding inclusive value:

$$V(b, z, h, a, 0, 0) = \kappa_F \log \sum_{\hat{n} \in \mathcal{N}} \exp\{\tilde{V}^F(\hat{n}|b, z, h, a, 0, 0)/\kappa_F\}. \quad (22)$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and  $V(b, z, h, a, n, s) = V^H(b, z, h, a, n, s)$ . The family-size decision therefore incorporates the housing position chosen in the same period. A birth can change current space and tenure immediately, not only the household's continuation value.

## 4.6 Stationary Equilibrium

Let  $g$  denote the beginning-of-period stationary composition of adult households over liquid wealth, productivity, inherited owned housing, age, family size, and child age,  $(b, z, h, a, n, s)$ , normalized to integrate to one. Total adult scale is  $S$ , so the corresponding level measure is  $G = Sg$ :

$$\int dg(b, z, h, a, n, s) = 1, \quad \int dG(b, z, h, a, n, s) = S. \quad (23)$$

The distinction between  $g$  and  $G$  separates composition from scale: household policies determine the composition of the population, while entry and replacement determine how many adults the modeled market contains.

**Definition 1** (Stationary equilibrium). A stationary equilibrium is a user cost and asset price  $(q, P)$ , value functions and policies, housing and fertility choice probabilities, stationary composition  $g$ , adult scale  $S$ , level measure  $G = Sg$ , and aggregate housing quantities such that:

1. households solve the branch problems (14)–(15), housing choice is given by (18), fertility choice is given by (21), and terminal utility is (5);
2. current choices, saving, income risk, child aging, adult aging, death, and the injection of new entrants induce a normalized transition operator  $\mathcal{T}(g; q, S)$ . Stationarity requires

$$g = \mathcal{T}(g; q, S); \quad (24)$$

3. let  $\Pi^H(h' \mid b, z, h, a, n, s)$  and  $\hat{h}^R(b, z, h, a, n, s)$  denote the housing-position probability and expected rental space conditional on renting, respectively, after integrating over the current family-size choice. Per-unit physical housing demand is

$$\begin{aligned} \hat{H}^D(g, q) = \int \left[ \Pi^H(0 \mid b, z, h, a, n, s) \hat{h}^R(b, z, h, a, n, s) \right. \\ \left. + \sum_{k=1}^K \Pi^H(H_k \mid b, z, h, a, n, s) H_k \right] dg(b, z, h, a, n, s). \end{aligned} \quad (25)$$

4. let  $E_0(g, q)$  be the inflow of young adults required to reproduce one unit of the stationary composition,  $B_0(g, q)$  the mature locally born flow per unit of adult scale,  $M$  the outside-origin entrant pool, and  $q^E(q)$  the probability that a potential entrant enters the modeled market. Entry balances adult replacement when

$$SE_0(g, q) = q^E(q) [M + SB_0(g, q)]. \quad (26)$$

Provided  $E_0(g, q) > q^E(q)B_0(g, q)$ , this condition gives the finite scale

$$S = \frac{q^E(q)M}{E_0(g, q) - q^E(q)B_0(g, q)}; \quad (27)$$

5. the housing market clears in levels and the user-cost relation holds:

$$S\hat{H}^D(g, q) = H^S(q), \quad q = (r + \delta + \tau^p)P; \quad (28)$$

6. taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.

Equilibrium links family formation to housing through two margins. Children change housing demand within a household, and, when locally born children feed the future entrant pool, they also change the scale of the market. Housing supply determines how much of either margin is absorbed by quantities rather than by the user cost.

The benchmark normalizes  $S = 1$ ; in counterfactuals, population is allowed to adjust. Computationally, for a candidate  $q$  we solve the household problems and the stationary composition, recover  $S$  from the entry condition, and update  $q$  until level housing demand equals supply. This nesting is a solution method, not a different equilibrium concept.

## 4.7 Computation

One model period is four years. Period  $j = 0, \dots, 16$  begins at age  $18 + 4j$ ; the last interval covers ages 82–85. Retirement begins at 66, and the last period in which a birth can occur begins at 42. Annual parameters are converted before solution:

$$1 + r = (1 + r_{\text{ann}})^4, \quad \beta = \beta_{\text{ann}}^4, \quad \delta = 1 - (1 - \delta_{\text{ann}})^4, \quad \tau^p = 4\tau_{\text{ann}}^p.$$

Income and consumption-flow quantities are expressed in four-year units; housing remains in rooms.

The search grid has  $N_b = 120$  nodes on  $[-12, 30]$ , with a dense core on  $[-5, 7]$  and a thinner upper tail beyond 15. Income takes values  $z \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$  with invariant weights  $\pi_z = (0.1, 0.2, 0.4, 0.2, 0.1)$  and transition matrix  $\Pi^z = 0.85I + 0.15\mathbf{1}\pi'_z$ . One dependent-child stage has an expected duration of 18 years, giving a four-year maturity probability of  $2/9$ . Entrants have mass  $1/J$  each period. The balanced pension equals payroll-tax revenue per retiree in the stationary age distribution.

Value functions are computed by backward induction. Golden-section search solves saving in each branch, and Howard iteration accelerates evaluation. The stationary distribution follows by forward iteration, while damped fixed-point iteration and a scalar bracketing refinement solve the housing market. I require a market residual below  $10^{-4}$  and exclude any solution that does not meet this threshold.

Because tenure and fertility choices are close to deterministic at the calibrated point, moments built on buy thresholds and the borrowing constraint move with the wealth grid, and grid resolution matters more here than in a standard lifecycle model. An earlier calibration at  $N_b = 60$  re-evaluated 18 percent worse at  $N_b = 120$  at the same parameters; part of its fit was grid noise. The July calibration was searched at  $N_b = 120$  and re-evaluated at  $N_b = 240$ . The re-estimation will use the same search and verification grids.

## 5 Calibration

### 5.1 Parameters Set Outside the Estimation

The model period is four years. I set standard financial parameters, the income process, and the housing menu outside the estimation. The entrant wealth distribution is taken directly from the PSID.

Table 1: Parameters set outside the estimation

| Parameter                       | Value                                             | Source or rule                                                                                                                    |
|---------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Interest rate $r$               | 4% per year                                       | standard                                                                                                                          |
| Depreciation $\delta$           | 2% per year                                       | standard                                                                                                                          |
| Property tax $\tau^p$           | 1% per year                                       | U.S. average effective rate                                                                                                       |
| Financed share $\phi$           | 0.80                                              | 20% down payment                                                                                                                  |
| Sale cost $\psi^s$              | 0.06                                              | transaction-cost evidence                                                                                                         |
| Risk aversion $\sigma$          | 2                                                 | standard                                                                                                                          |
| Bequest shift $\theta_1$        | 0.01                                              | near-linear warm glow                                                                                                             |
| Payroll tax $\tau_{\text{pay}}$ | 0.179                                             | pension balances in the stationary distribution                                                                                   |
| Income profile $e_a$            | 0.65 at 18, rises to 1.0 at 34, declines after 58 | normalized lifecycle profile; five-state persistent income process                                                                |
| Owner menu $\{H_k\}$            | $\{2, 4, 6, 8, 10\}$ rooms                        | room-count support, ACS                                                                                                           |
| Renter cap $\bar{h}^R$          | 6.0 rooms                                         | 90th-percentile rental size rule of <a href="#">Greaney et al. (2025)</a> , measured in the matched ACS sample; 6.5 as robustness |
| Supply $(\bar{H}, \eta)$        | (4, 1)                                            | unit-elastic benchmark                                                                                                            |
| Entry wealth                    | PSID young-renter wealth-to-income distribution   | first stage                                                                                                                       |

The renter cap is the quantitative counterpart of  $\bar{h}^R$  in the analytical model. I measure it from the rental-size distribution rather than estimate it internally. The fit is nearly flat between 6.0 and 6.5 rooms and deteriorates outside that range. I likewise use the support of rooms in the ACS to choose the owner ladder. These choices describe the available housing products; the estimation determines how households value them and sort across them.

## 5.2 Estimated parameters

I estimate the remaining 13 parameters by matching 14 moments on family size, housing consumption, tenure, and lifecycle wealth. The estimates minimize a weighted distance between the model and the data using  $N_b = 120$  points for liquid wealth. Fourteen moments provide one overidentifying restriction by count. In practice, however, several parameters have similar effects on the moments, so the estimates should not all be interpreted as precisely identified.<sup>8</sup>

<sup>8</sup>I first search globally using differential evolution and then refine the best candidates with local minimization.



Table 2: Internally estimated parameters (July 9 calibration)

| Parameter               | Interpretation                                | Estimate |
|-------------------------|-----------------------------------------------|----------|
| $\beta_{\text{annual}}$ | Annual discount factor                        | 0.940    |
| $\alpha_c$              | Consumption weight                            | 0.620    |
| $\bar{c}_0$             | Baseline consumption requirement              | 1.279    |
| $\bar{c}_n$             | Consumption cost per dependent child          | 0.116    |
| $\bar{h}_0$             | Baseline housing requirement                  | 1.000    |
| $\bar{h}_{\text{jump}}$ | Discrete housing-need jump at the first child | 2.120    |
| $\bar{h}_n$             | Housing requirement per dependent child       | 1.382    |
| $\psi_{\text{child}}$   | Utility from children                         | 0.349    |
| $\kappa_F$              | Family-size choice scale                      | 1.019    |
| $\kappa_T$              | Tenure-choice scale                           | 0.000    |
| $\chi_O$                | Owner housing-service multiplier              | 1.150    |
| $\theta_0$              | Bequest strength                              | 0.0015   |
| $\theta_n$              | Bequest shifter by number of children         | 0.972    |

No parameter is identified by a single moment. I describe the main sources of identification below.

Lifecycle wealth and homeownership profiles identify the saving parameters. The discount factor  $\beta$  primarily changes asset accumulation, while the baseline consumption requirement  $\bar{c}_0$  reduces the resources available for saving, housing, and children. I use liquid wealth among young renters, old-age nonhousing wealth, and homeownership at young and old ages to separate these effects. The separation remains weak because both parameters influence tenure through accumulated wealth.

I identify housing needs from how housing consumption changes with family composition. Mean rooms among childless renters identifies the baseline requirement  $\bar{h}_0$ ; the change in rooms when the first child arrives identifies  $\bar{h}_{\text{jump}}$ ; and the difference in rooms between larger and smaller families identifies the additional requirement  $\bar{h}_n$ . These moments also respond to income, wealth, and tenure, so the three estimates are strongly correlated.

The consumption weight  $\alpha_c$  changes housing demand in both tenure arrangements, while  $\chi_O$  changes the value of owner housing relative to rental housing. I use the owner–renter room difference, the share of owners in large homes, and homeownership by age and family status to distinguish the two. The distinction is imperfect because wealth and borrowing constraints also affect whether a household owns.

Completed fertility and childlessness identify the consumption cost of children  $\bar{c}_n$ , the utility value of children  $\psi_{\text{child}}$ , and the dispersion of family-size choices  $\kappa_F$ . The three parameters have different economic roles:  $\bar{c}_n$  reduces the resources left for other goods,  $\psi_{\text{child}}$  raises the direct value of children, and  $\kappa_F$  determines how dispersed choices are across otherwise similar households. Nevertheless, these effects can substitute for one another locally. Homeownership by age and family status identifies  $\kappa_T$ . Its estimate of zero means that additional random variation in tenure choice does not improve the fit; it does not mean that the parameter is precisely estimated.

Finally, I use old-age wealth and homeownership to identify the overall strength of the bequest motive  $\theta_0$ , and the parent–childless difference in old-age wealth to identify how that motive varies with family size through  $\theta_n$ . The second effect is weakly identified because the estimated overall bequest motive is close to zero. I therefore treat the estimates as a provisional parameterization, not as evidence that every preference parameter is tightly identified.<sup>9</sup>

<sup>9</sup>Several estimates lie at or near the limits of the July search region.

### 5.3 Target moments and fit

The calibration targets fourteen moments: completed fertility and childlessness from the CPS (the June 2024 supplement cells for women 40–44), tenure and housing-size moments by age and family status from a matched ACS metropolitan sample, and wealth moments from the PSID. Table 3 compares the data with the model at the July 9 parameterization. The weighted objective is 6.31.

Table 3: July 9 target moments and model fit ( $N_b = 120$ ; provisional loss 6.31)

| Moment                                                | Data  | Model |
|-------------------------------------------------------|-------|-------|
| Old-age ownership rate (65–75)                        | 0.764 | 0.893 |
| Ownership rate (25–34)                                | 0.341 | 0.226 |
| Owner–renter mean rooms (no children at home, 30–55)  | 2.419 | 2.151 |
| Family ownership gap                                  | 0.168 | 0.298 |
| Young renter liquid wealth / income (25–35)           | 0.179 | 0.353 |
| Ownership rate (30–55)                                | 0.575 | 0.531 |
| Renter mean rooms (no children at home, 30–55)        | 3.805 | 3.655 |
| Childlessness rate                                    | 0.188 | 0.260 |
| Rooms gap, 3+ vs. 1–2 children (30–55)                | 0.368 | 0.269 |
| Old parent–childless nonhousing wealth gap            | 1.007 | 0.842 |
| Housing increment at first birth (rooms)              | 0.664 | 0.618 |
| Completed fertility                                   | 1.918 | 1.949 |
| Old nonhousing wealth / income, median (65–75)        | 2.230 | 2.328 |
| Owner share with rooms $\geq 6$ (no children at home) | 0.596 | 0.593 |

The July calibration is close to the fertility level and several size-distribution moments. Old-age ownership is too high: the model keeps 89 percent of 65–75 households in owned housing against 76 percent in the data, because nothing in the model pushes an old owner out of her house. There are no health shocks and no mortality-linked downsizing, so retention is close to absorbing. Ownership at 25–34 is too low (0.23 against 0.34), while young renter wealth is too high (0.35 against 0.18). Both moments are shaped by accumulated wealth and access to ownership and must be interpreted jointly. Childlessness also overshoots (0.26 against 0.19), even though average completed fertility is close to the data. That combination points to a miss in the distribution of family size rather than its mean.

This table reports the calibration before correcting purchase timing. Correcting the timing of current housing choices changes several tenure and housing moments, including the same-period housing response at the first birth. I therefore use the table to describe the provisional fit, not as the baseline for the policy results.

Re-evaluated at  $N_b = 240$  at the same parameters, the loss rises from 6.31 to 6.84, and the difference concentrates in the family ownership gap and the old-age ownership rate, both of which move further from their targets at the finer grid. I will report the same 120/240 comparison after re-estimation.

## 6 Lifecycle Housing Patterns

The preliminary calibration produces a steep rise in homeownership near the end of the fertile period and high homeownership at older ages. Figure 3 should be interpreted cautiously because purchases are measured one model period late in this version. The new calibration will update the profile.

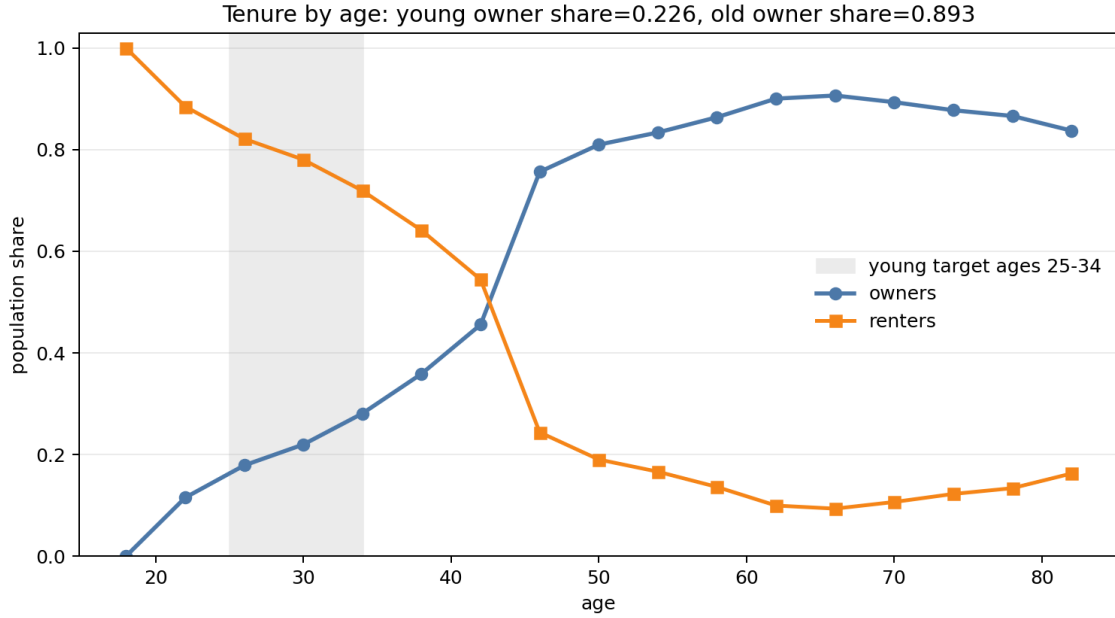


Figure 3: Ownership over the lifecycle in the preliminary calibration. The steep rise near the end of the fertile window and high old-age ownership are visible, but current purchases are recorded one model period late in this version.

Families that rent often choose the largest available rental unit, while owners choose among a discrete ladder of house sizes. Many young renters also hold little liquid wealth. Together, these patterns show how rental scarcity and down-payment constraints can bind at ages when households are forming families. Their quantitative prevalence will be reassessed after re-estimation.

## 7 Quantitative Evidence on the Mechanism

The analytical model predicts that a constrained household values an additional room by more than its market user cost. I denote this difference by  $\zeta$  and measure it in the current calibration. Economically, it answers a simple question: how much more would a household pay for another room if the rental-size or financing constraint were relaxed?

Among fertile-age renters choosing the largest available rental unit, the average value of  $\zeta$  is 0.248 goods units, or 1.28 times the market user cost. These households therefore value another room at about 2.3 times its market price. When I weight households by their predicted family-size response, 73 percent of the aggregate response comes from households with  $\zeta > 0$ . This share is similar across the grid resolutions and owner menus I have considered, but it remains preliminary because correcting purchase timing changes which households are exposed to the constraint.

The model places many purchases near the fertility deadline. The purchase region expands in the last fertile period, when the option value of waiting to have children expires. Buyers in this region hold wealth above the required down payment. Housing constraints therefore do more than force households to save toward a distant threshold: they make the purchase and family-size decisions occur together near the deadline. I will revisit this pattern after re-estimation.

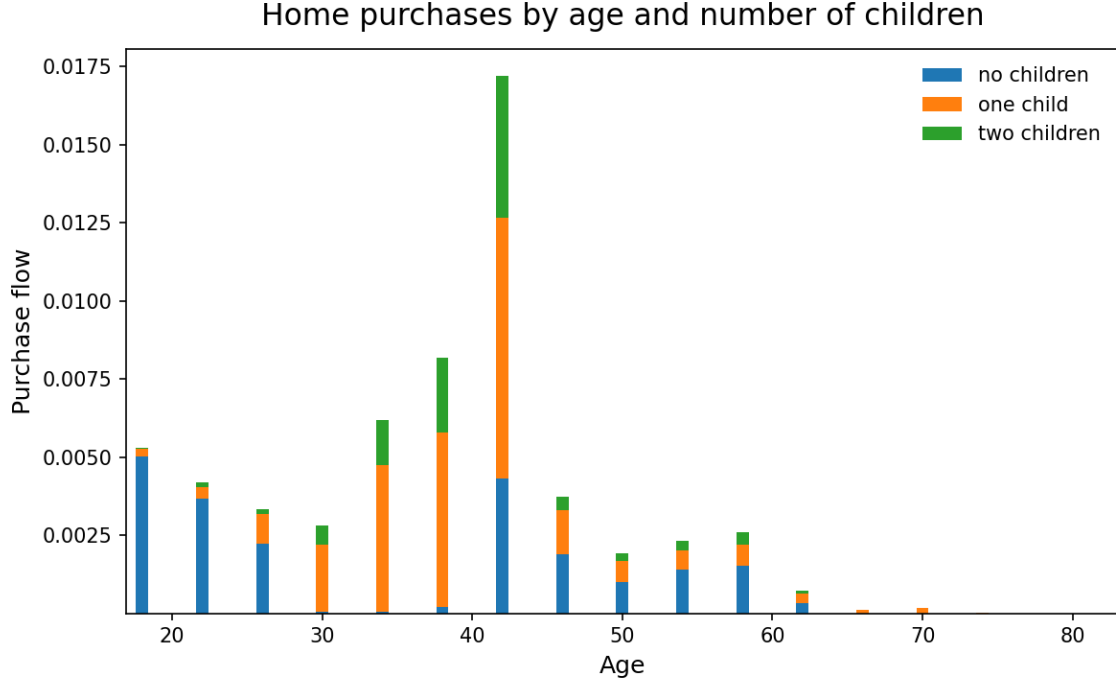


Figure 4: Purchase flows by age and family size in the July 9 calibration. Purchases concentrate near the end of the fertile window and often coincide with a birth.

The model is motivated by the allocation of large homes across generations. The next table compares who owns homes with at least six rooms in the current calibration and in the ACS.

Table 4: Who holds the family-sized owner stock (rooms  $\geq 6$ )

| Holder                                      | Data (ACS) | Model |
|---------------------------------------------|------------|-------|
| Age 18–29                                   | 3.4%       | 0.4%  |
| Age 30–45                                   | 24.8%      | 12.7% |
| Age 46–64                                   | 43.3%      | 50.5% |
| Age 65+                                     | 28.5%      | 36.5% |
| Held by households 46+                      | 71.8%      | 87.0% |
| Held by households with no children at home | 54.9%      | 72.5% |

The concentration is present in the data, not only in the model. In the ACS, 72 percent of family-sized owner housing is held by households past the fertile years and 55 percent by households with no children at home. The model has the same ordering but overstates the concentration among older households. I will recompute the model column after re-estimation; the data column already shows the allocation problem that motivates the paper.

I also use the model to study a direct incentive for older owners to release housing. In the current calibration, this experiment changes the housing price by less than one percent and completed fertility by about one hundredth. The small response does not show that old-owner retention is unimportant. It reflects the absence of health shocks, mortality risk, and other reasons to downsize: most old owners are already at a keep-everything corner, so few can respond to the policy. I therefore

view this result as preliminary.

## 8 Housing Policy

I next conduct a preliminary study of housing policies. The idea is to ask whether policies that move family-sized space toward households that value it more can reduce the inefficiency identified in the analytical model, and whether this reallocation changes family formation. I consider a holding tax on large owner-occupied homes, a birth-linked down-payment grant restricted to family-sized homes, greater access to large rental units, a direct incentive for older owners to release housing, and a package that combines the tax and the grant. These are exploratory exercises based on the July calibration, not policy estimates. Housing prices adjust to clear the market in each case, and I will recompute all of the results after re-estimation.

I begin with a two percent holding tax on large owner-occupied homes. Its main effect is capitalization: the asset price of these homes falls by 12 percent, which lowers the down payment required to buy them. The price response is similar across the alternative grids, moment weights, and owner menus that I have checked, but completed fertility rises by only +0.004 to +0.009. The response is small because the households affected by the tax already hold wealth above the purchase threshold. A lower price changes which house they buy more than whether they have children.

I then consider a down-payment grant paid at a birth and restricted to family-sized homes. Completed fertility rises by +0.01, +0.13, +0.26, and +0.49 at grant sizes 0.1, 0.4, 0.58, and 1.0, measured in units of mean annual income. The grant acts more directly than the holding tax because it relaxes the financing constraint when the household is deciding whether to have children. Conditioning the grant on a birth also creates a distortion: some households delay a purchase to qualify for the subsidy. Young ownership falls from 0.23 to 0.09 at the smallest grant. This delay fades at larger grant sizes, but it is an important limitation of a birth-contingent policy.

Finally, I combine the two policies. The holding tax reduces the value of keeping a large home and lowers its price, while the grant helps young households finance the purchase. In the July calibration, the two percent tax funds the grant, lowers the price of large homes by 11 percent, and raises completed fertility by +0.15. This magnitude is provisional.

### 8.1 Market Entry and Population

The results above hold adult population fixed at  $S = 1$  to isolate changes in the behavior of households already in the market. I next allow the policies to change market size. A policy that makes the market more attractive draws in more households; if it changes births, it also changes the future flow of locally born entrants. Equation (26) determines the resulting population, and equation (28) determines the new housing price. I choose the outside value and entrant pool to reproduce  $S = 1$  in the benchmark and hold them fixed across the policy exercises. The calculation assumes that locally born children become future potential entrants one-for-one.

For comparability, I solve all cases at  $N_b = 120$  using the July calibration. The grant and package results retain a few-percent price imprecision and should therefore be read as preliminary.

Table 5: Preliminary policy exercises with endogenous entry ( $N_b = 120$ )

| Policy                    | $\Delta\text{TFR}$ , fixed pop. | $\Delta\text{TFR}$ , intensive | Scale $S$ | Total births    |
|---------------------------|---------------------------------|--------------------------------|-----------|-----------------|
| Holding tax 2%            | +0.004                          | +0.002                         | 1.003     | +0.4%           |
| Grant 0.4, rungs $\geq 6$ | +0.119                          | +0.083                         | 1.038     | +8.2%           |
| Package (tax + grant)     | +0.162                          | +0.086                         | 1.040     | +8.6%           |
| Rental cap 6.5            | -0.006                          | -0.009                         | 1.004     | $\approx 0.0\%$ |
| Retention wedge 0.2       | +0.006                          | +0.000                         | 1.007     | +0.7%           |

Allowing entry attenuates the effect of the grant on families already in the market. The grant attracts more households, population expands, and the resulting increase in the housing price offsets part of the fixed-population fertility response. Total births nevertheless rise in the grant and package exercises. The holding tax and the direct incentive to older owners remain close to neutral. The one-market model cannot determine whether the increase in local births is creation or relocation from elsewhere.

## 9 Measurement and Re-estimation

The July estimates measure some housing choices with a one-period lag. In the household problem, a buyer receives the new house immediately, but the aggregation step used inherited housing to compute market demand and several empirical moments. I have corrected the aggregation so that a current purchase contributes to current housing demand and to the measured housing response when the first child arrives. Because this changes the moments the model is asked to match, the parameter vector must be re-estimated.

I have also reconciled the income profile, wealth grid, housing menu, and target definitions used for the July estimates with the maintained specification. The new estimation will use 120 wealth points, and I will check the preferred estimates on a 240-point grid before recomputing the policy results. The written model now describes the preferences and financial constraints used in the July estimation: the owner premium applies to usable space  $H_k - \bar{h}(n, s)$ , bequest utility varies with family size, and all owners face the same debt limit.

## 10 Limitations

The model remains deliberately incomplete after these repairs. Old-age ownership is too high because health, mortality, and other reasons for downsizing are absent. Fertility has no age-dependent fecundity, so waiting is costless until the final fertile period. Both omissions matter for the timing and stock-allocation questions at the center of the paper.

In the quantitative model, transaction costs are the only reason older owners are reluctant to sell. It does not yet include the capital-gains and basis mechanism of Section 3. Housing supply is unit-elastic, whereas the analytical reallocation result holds the stock fixed. Finally, the entry results depend on the benchmark outside option and on the assumption that local births become future entrants. They should therefore be read as a decomposition, not as an estimated migration response.

## 11 Conclusion

Children use housing space, and access to that space differs sharply by tenure. A rental-size limit raises the shadow price of room for a constrained household. Ownership relaxes the size limit but introduces a down payment. Retention frictions can keep large homes with older households whose current use of space is low. The analytical model shows that these constraints raise the effective cost of accommodating children and can leave family-sized space with households that value it less.

The preliminary calibration suggests that this mechanism is active. Many households near the fertility margin value additional space above its user cost, purchases bunch late in the fertile window, and the data show that older households hold most family-sized owner housing. The size of these effects is not yet settled because correcting purchase timing changes several moments used in estimation. The new calibration will show whether the importance of housing constraints, and the policy responses they generate, survive that correction. The central quantitative question is whether reallocating family-sized homes can meaningfully change family size and total births.

## A Additional Analytical Results

This appendix records three extensions to the short analytical note. They are not needed for the main argument in Section 3.

First, at equal scaling,  $\chi = (1 + \Lambda)\kappa q^m/\alpha$ , the fixed-mode response to a marginal relaxation of a binding housing cap can be written as

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m - \chi n_i^m} \omega_i^m, \quad \omega_i^m > 0.$$

The normalized access gap therefore governs both the response of family size among incumbent households and the response of entry into the modeled market.

Second, for a collateral-capped owner,  $H_i^O = b_i \rho / [(1 - \phi)q]$ , with  $\rho = r + \delta + \tau^p$ . Holding the user cost and the old-owner gains wedge fixed,

$$\left. \frac{\partial H_i^O}{\partial \tau^p} \right|_{q, \bar{\ell}} = \frac{b_i}{(1 - \phi)q} > 0. \quad (29)$$

A property-tax increase can therefore relax the down-payment constraint by lowering the capitalized asset price, even when the housing user cost is held fixed.

Finally, a policy can move households across the rent-own boundary as well as relax a constraint within a tenure. The resulting change in births depends on both the mass of households near that boundary and the difference in family size between their owner and renter choices. Its sign is not determined by the direction of tenure switching alone.

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